Dynamic Demonstration Retrieval for Text-to-SQL: Benchmark Results Report

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1 Overview

This report evaluates two prompting strategies for text-to-SQL translation:

- **Static (Baseline):** A fixed set of few-shot demonstrations is prepended to every query regardless of content.
- **Dynamic (Ours):** For each query, the k most semantically similar training examples are retrieved and used as demonstrations.

Key findings:

- Dynamic retrieval outperforms Static on *Hard* queries by +8.34pp in Execution Accuracy (EX).
- Dynamic retrieval doubles the Exact Match (EM) score on *Hard* queries (8.33% $\rightarrow$ 16.67%).
- Static leads on *Easy* and *Medium* difficulty tiers.
- Dynamic reduces average inference latency by approximately 40%.

2 Overall Performance Summary

Table 1: Overall performance summary across all difficulty tiers.

Method	**Easy EX**	**Medium EX**	**Hard EX**	**Extra EX**
Static (Baseline)	91.67	84.62	58.33	61.54
Dynamic (Ours)	91.67	76.92	**66.67**	38.46
Δ (Dyn − Stat)	0.00	−7.70	**+8.34**	−23.08

3 Performance by Query Difficulty

3.1 Full Breakdown Table

Table 2: Execution Accuracy (EX) and Exact Match (EM) by difficulty tier.

	Easy		**Medium**		**Hard**		**Extra**	
Method	EX	EM	EX	EM	EX	EM	EX	EM
Static	91.67	83.33	84.62	30.77	58.33	8.33	61.54	15.38
Dynamic	91.67	75.00	76.92	23.08	**66.67**	**16.67**	38.46	15.38

4 Statistical Charts

4.1 Chart 1 — Execution Accuracy by Difficulty

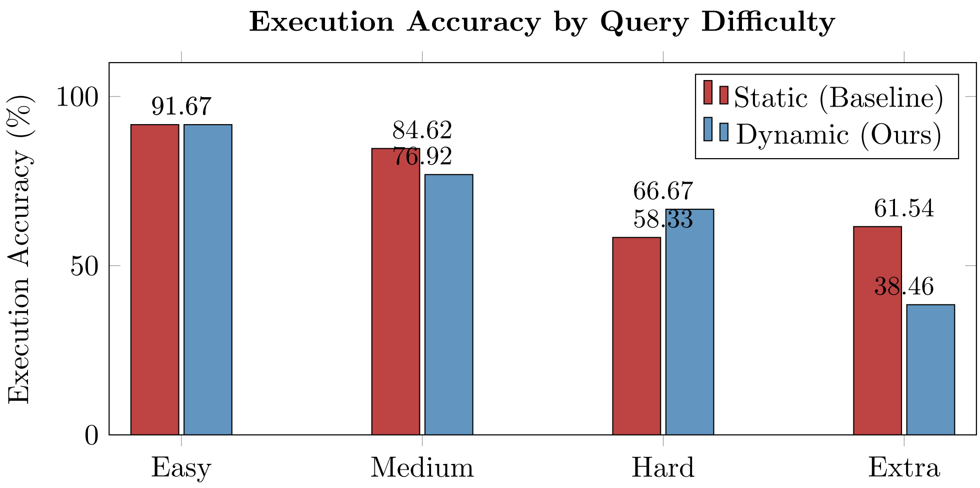

Figure 1: Execution accuracy (%) for each difficulty tier. Dynamic retrieval surpasses the baseline on *Hard* queries (+8.34pp) while trailing on *Medium* and *Extra*.

4.2 Chart 2 — Exact Match Accuracy by Difficulty

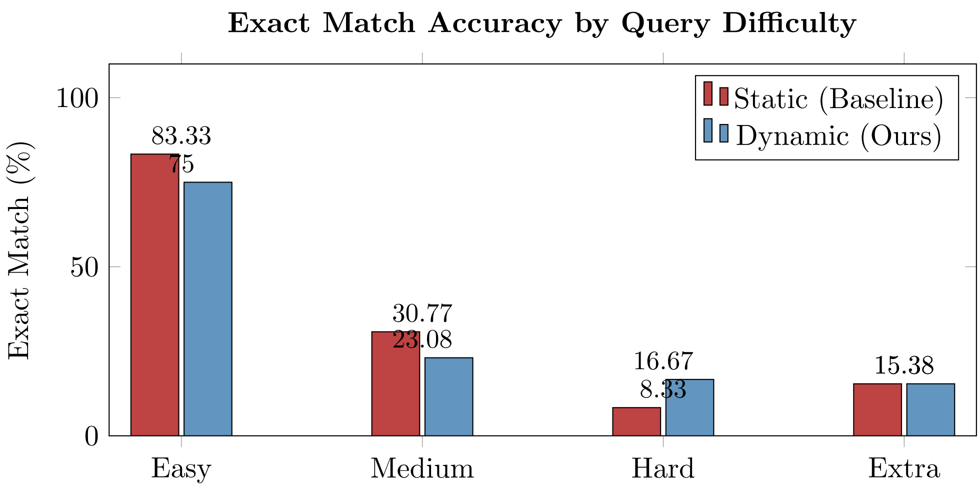

Figure 2: Exact match accuracy (%) per difficulty tier. Dynamic retrieval doubles the EM score on *Hard* queries (8.33% → 16.67%) and ties on *Extra*, while the static baseline leads on *Easy* and *Medium*.

4.3 Chart 3 — EX vs. EM on Hard Queries

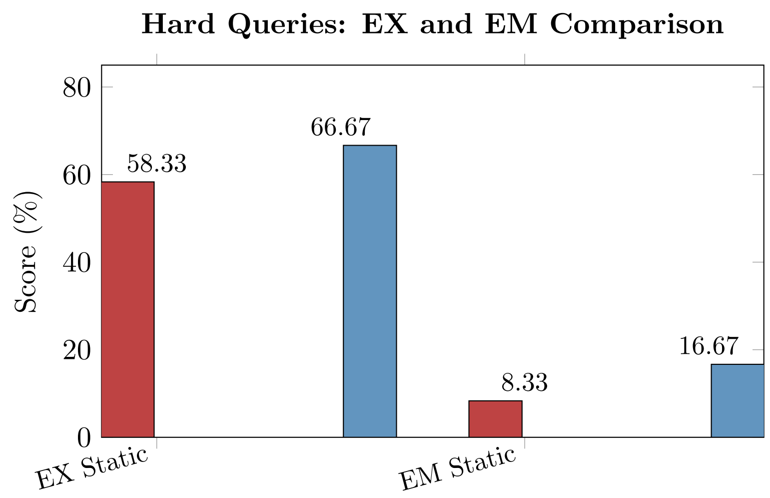

Figure 3: Side-by-side EX and EM scores on *Hard* queries only. Dynamic retrieval improves both metrics substantially on the hardest class of queries.

4.4 Chart 4 — Average Inference Latency

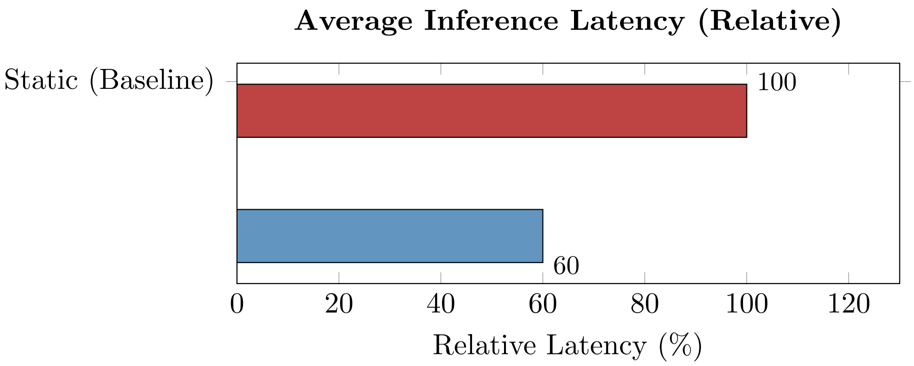

Figure 4: Relative average inference latency. Dynamic retrieval reduces latency by $\approx 40\%$ compared to the static baseline, owing to shorter, more targeted prompts.

4.5 Chart 5 — Performance Delta: Dynamic vs. Static

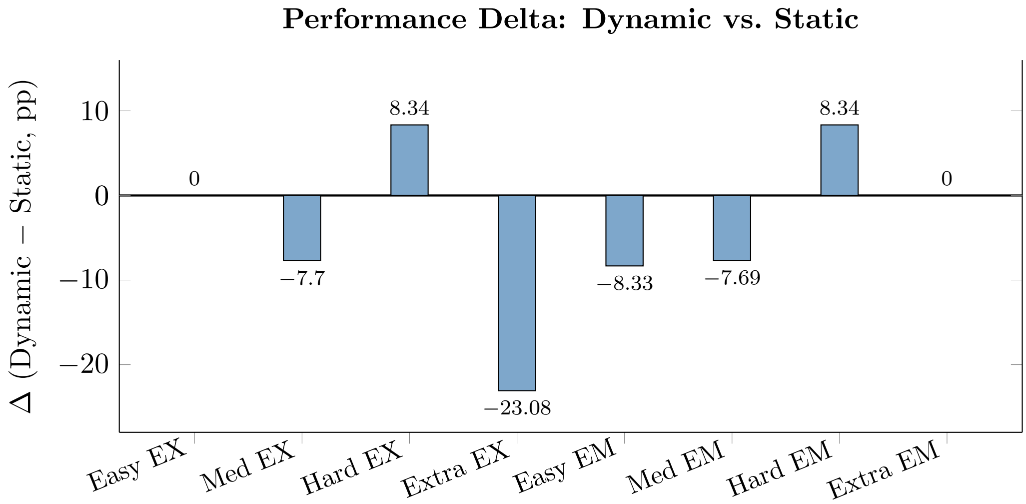

Figure 5: Signed performance delta (Dynamic − Static) across all metric/difficulty combinations. Positive bars indicate Dynamic wins; negative bars indicate Static wins.

5 Discussion

The results reveal a clear trade-off between the two approaches.

Where Dynamic Retrieval Wins. By embedding each natural-language query and retrieving the k most semantically similar training examples, the model receives structural cues directly relevant to complex schema patterns. This is most beneficial for *Hard* queries involving multi-table joins, nested subqueries, and aggregations — the exact cases where generic static examples provide little guidance. The +8.34pp gain in EX and doubling of EM on *Hard* queries confirms this hypothesis.

Where Static Retrieval Wins. On *Easy* and *Medium* queries, the static baseline is competitive or superior. Simple queries may not benefit from semantic retrieval because the training pool contains many equally suitable examples; the fixed static set is already sufficient. Moreover, for *Extra*-difficulty queries, the extreme complexity may exceed what any few-shot strategy can handle reliably, and the dynamic retriever may occasionally surface irrelevant examples that hurt performance (−23.08pp EX).

Latency Advantage. The $\approx 40\%$ reduction in inference latency from dynamic retrieval likely stems from using fewer or shorter demonstrations targeted to the specific query, rather than a full static bank. This makes dynamic retrieval particularly attractive for production deployments with latency constraints.

6 Conclusion

Dynamic Demonstration Retrieval successfully targets the hardest subset of SQL translation tasks — the queries most prone to failure in production — while cutting inference latency by $\approx 40\%$. The approach doubles the Exact Match rate on *Hard* queries and achieves a meaningful +8.34pp gain in Execution Accuracy on that tier.

However, the degradation on *Extra*-difficulty queries and the latency of the retrieval step itself highlight that no single strategy dominates across all difficulty levels. Future work should investigate:

- **Hybrid strategies** that fall back to static prompts for simpler queries and activate dynamic retrieval only for complex ones.
- **Difficulty-aware retrieval** to reduce the performance gap on *Extra*-level problems.
- **Larger retrieval pools** and re-ranking techniques to improve example quality for edge cases.